# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 10 — SCIENCE · NCERT-TYPE MCQ TEST SET 2

**Full Syllabus (Chapters 1–13)** | RBSE Pattern | Session 2026–27

**Time: 1 Hour** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. NCERT-type: questions in the style of the textbook examples and exercises.

## MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

*Choose the correct option and write it in your answer book.*

**1.** Magnetic field lines inside a bar magnet go from 1

**(a)** the north pole to the south pole **(b)** the south pole to the north pole
**(c)** one side to the other randomly **(d)** do not exist

**2.** Which of the following is an abiotic component of an ecosystem? 1

**(a)** Fungi **(b)** Temperature **(c)** Bacteria **(d)** Grass

**3.** The white precipitate formed on mixing sodium sulphate and barium chloride solutions is: 1

**(a)** $BaCl_2$ **(b)** $BaSO_4$ **(c)** $NaCl$ **(d)** $Na_2SO_4$

**4.** Tooth decay starts when the pH of the mouth falls below: 1

**(a)** 3 **(b)** 5.5 **(c)** 9 **(d)** 7

**5.** A metal that reacts with cold water to produce hydrogen but floats because of bubbles of hydrogen: 1

**(a)** sodium **(b)** magnesium **(c)** potassium **(d)** calcium

**6.** The cleansing action of soap is due to: 1

**(a)** its acidity **(b)** formation of micelles
**(c)** reaction with dirt **(d)** its smell

**7.** In muscles during vigorous exercise, glucose breaks down anaerobically to give: 1

**(a)** ethanol **(b)** $CO_2$ only **(c)** water **(d)** lactic acid

**8.** The roots of a plant grow downward in response to: 1

**(a)** air **(b)** light **(c)** gravity **(d)** water only

**9.** The part of the flower that becomes the fruit is the 1

**(a)** ovule **(b)** ovary **(c)** stigma **(d)** style

**10.** The F2 phenotypic ratio in the dihybrid cross is 1

**(a)** 3 : 1 **(b)** 1 : 2 : 1 **(c)** 9 : 3 : 3 : 1 **(d)** 1 : 1 : 1 : 1

**11.** When light goes from air into glass, it 1

**(a)** bends away from the normal **(b)** bends towards the normal
**(c)** does not bend **(d)** gets reflected completely

**12.** Hypermetropia is corrected by using 1

**(a)** a concave lens **(b)** a convex lens **(c)** a concave mirror **(d)** a prism

**13.** The SI unit of resistance is 1

**(a)** ohm **(b)** ampere **(c)** watt **(d)** volt

**14.** A soft iron core inside a solenoid 1

**(a)** reduces the field **(b)** makes an electromagnet

**(c)** stops the current **(d)** has no effect

**15.** Which of the following is a correct food chain? 1

**(a)** Grass, grasshopper, frog, snake **(b)** Snake, frog, grasshopper, grass

**(c)** Grass, snake, frog, grasshopper **(d)** Frog, grass, snake, grasshopper

**16.** After balancing $H_2 + Cl_2 \longrightarrow HCl$, the coefficient of HCl is: 1

**(a)** 1 **(b)** 2 **(c)** 4 **(d)** 3

**17.** Phenolphthalein turns pink in: 1

**(a)** basic solution **(b)** any solution **(c)** acidic solution **(d)** neutral solution

**18.** In the electrolytic refining of copper, the anode is made of: 1

**(a)** impure copper **(b)** graphite **(c)** pure copper **(d)** zinc

**19.** Carbon forms covalent bonds because it has: 1

**(a)** six valence electrons **(b)** two valence electrons

**(c)** eight valence electrons **(d)** four valence electrons

**20.** Urine is stored temporarily in: 1

**(a)** kidney **(b)** urethra **(c)** urinary bladder **(d)** ureter

**21.** The master gland of the body is the: 1

**(a)** thyroid **(b)** pancreas **(c)** pituitary **(d)** adrenal

**22.** The testes lie outside the abdominal cavity in the scrotum because 1

**(a)** they need more oxygen

**(b)** they produce urine

**(c)** they are too large for the abdomen

**(d)** sperm formation needs a temperature lower than body temperature

**23.** A normal human female has the sex chromosomes 1

**(a)** YY **(b)** XO **(c)** XY **(d)** XX

**24.** The power of a lens of focal length 50 cm is 1

**(a)** +2 D **(b)** -2 D **(c)** +0.5 D **(d)** +5 D

**25.** Which colour bends least on passing through a glass prism? 1

**(a)** Violet **(b)** Blue **(c)** Green **(d)** Red

**26.** In a series circuit the quantity that is the same through all the resistors is 1

**(a)** power **(b)** potential difference

**(c)** current **(d)** resistance

**27.** The phenomenon of electromagnetic induction was discovered by 1

**(a)** Newton **(b)** Michael Faraday **(c)** Oersted **(d)** Ampere

**28.** Ozone is a molecule made of 1

**(a)** three oxygen atoms **(b)** oxygen and nitrogen

**(c)** two oxygen atoms **(d)** one oxygen atom

**29.** A solution of substance X is used for whitewashing. X is: 1

**(a)** Calcium carbonate **(b)** Calcium sulphate
**(c)** Sodium hydroxide **(d)** Calcium oxide (quick lime)

**30.** 10 mL of NaOH solution is completely neutralised by 8 mL of HCl. If 20 mL of the same NaOH is taken, the HCl needed is: 1

**(a)** 4 mL **(b)** 16 mL **(c)** 8 mL **(d)** 12 mL

**31.** An element reacts with oxygen to give a compound with a high melting point, soluble in water. The element is likely to be: 1

**(a)** iron **(b)** silicon **(c)** calcium **(d)** carbon

**32.** Which of the following is an unsaturated hydrocarbon? 1

**(a)** $CH_4$ **(b)** $C_3H_8$ **(c)** $C_2H_6$ **(d)** $C_2H_4$

**33.** Which of the following are required for autotrophic nutrition in green plants? 1

**(a)** only chlorophyll and sunlight
**(b)** carbon dioxide, water, chlorophyll and sunlight
**(c)** only sunlight
**(d)** only carbon dioxide and water

**34.** Which of the following is not a plant hormone? 1

**(a)** insulin **(b)** cytokinin **(c)** auxin **(d)** gibberellin

**35.** Hydra reproduces asexually mainly by 1

**(a)** multiple fission **(b)** binary fission **(c)** spore formation **(d)** budding

**36.** The transmission of characters from parents to offspring is called 1

**(a)** regeneration **(b)** evolution **(c)** heredity **(d)** variation

**37.** The centre of the reflecting surface of a spherical mirror is the 1

**(a)** centre of curvature **(b)** principal focus
**(c)** aperture **(d)** pole

**38.** The size of the pupil is controlled by the 1

**(a)** retina **(b)** ciliary muscles **(c)** optic nerve **(d)** iris

**39.** Electric current is measured by an 1

**(a)** ammeter connected in series **(b)** ammeter connected in parallel
**(c)** voltmeter connected in series **(d)** voltmeter connected in parallel

**40.** Two magnetic field lines 1

**(a)** intersect often **(b)** intersect at the centre
**(c)** intersect at the poles **(d)** never intersect each other

**41.** Which of the following are producers? 1

**(a)** Snakes **(b)** Fungi **(c)** Deer **(d)** Green plants

**42.** Oxidation is the: 1

**(a)** loss of oxygen or gain of hydrogen **(b)** gain of oxygen or loss of hydrogen
**(c)** gain of both oxygen and hydrogen **(d)** loss of both oxygen and hydrogen

**43.** The acid present in tamarind is: 1

**(a)** tartaric acid **(b)** citric acid **(c)** oxalic acid **(d)** lactic acid

**44.** Which of the following metals does not react with dilute HCl? 1

**(a)** magnesium **(b)** copper **(c)** zinc **(d)** iron

**45.** Soap does not lather well in hard water because of: 1

**(a)** calcium and magnesium salts forming scum **(b)** sodium chloride

**(c)** acidity **(d)** low temperature

**46.** In yeast, anaerobic respiration produces: 1

**(a)** ethanol and carbon dioxide **(b)** water

**(c)** glucose **(d)** lactic acid

**47.** Tendrils coil around a support due to: 1

**(a)** thigmotropism **(b)** geotropism **(c)** chemotropism **(d)** phototropism

**48.** The ovule develops into a 1

**(a)** petal **(b)** seed **(c)** fruit **(d)** flower

**49.** In the dihybrid F2, the number of round green seeds out of 16 is 1

**(a)** 9 **(b)** 3 **(c)** 4 **(d)** 1

**50.** When light passes from a denser to a rarer medium, it 1

**(a)** bends away from the normal **(b)** goes straight

**(c)** stops **(d)** bends towards the normal

**END OF QUESTION PAPER** **Total: 50 Marks**

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